fn gamma_lo_from_x

Michael Shoemate

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Proves soundness of `fn gamma_lo_from_x` in `mod.rs`.

1 Hoare Triple

Precondition

`x` is finite.

Pseudocode

```
1 def gamma_lo_from_x(x: float):
2     expm1_hi = (-x).inf_exp_m1()  #
3     return -expm1_hi
```

Postcondition

Theorem 1.1. For any setting of the input parameter `x` such that the given precondition holds, `gamma_lo_from_x` either returns `Err(e)`, or returns `Ok(out)` where

$$\texttt{out} \leq 1 - e^{-\texttt{x}}.$$

2 Proof

Assume the precondition is met.

Lemma 2.1. `gamma_lo_from_x` only returns `Err(e)` if the call to `inf_exp_m1` on line 2 returns `Err(e)`.

Proof. The only fallible operation in `gamma_lo_from_x` is the invocation of `(-x).inf_exp_m1()` on line 2. Negation of a finite floating-point value is infallible, and negating the result on the return line is infallible as well. Therefore every error returned by `gamma_lo_from_x` must come from `inf_exp_m1`. □

Lemma 2.2. If the outcome of `gamma_lo_from_x` is `Ok(out)`, then

$$\texttt{out} \leq 1 - e^{-\texttt{x}}.$$

Proof. Suppose `gamma_lo_from_x` returns `Ok(out)`. Write `expm1_hi` for the value returned by `(-x).inf_exp_m1()` on line 2. By the proof definition of `inf_exp_m1`,

$$\texttt{expm1_hi} \geq e^{-\texttt{x}} - 1.$$

Multiplying both sides by -1 reverses the inequality, giving

$$-\texttt{expm1_hi} \leq 1 - e^{-\texttt{x}}.$$

By the return statement of the function, $\texttt{out} = -\texttt{expm1_hi}$. Therefore

$$\texttt{out} \leq 1 - e^{-\texttt{x}},$$

as claimed. $\square$

Proof of Theorem 1.1. Holds by Lemma 2.1 and Lemma 2.2. $\square$